

SAIFULLA

PCB Design & Embedded Hardware Engineer

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PROFESSIONAL SUMMARY

Embedded Systems Engineer with hands-on experience in PCB design (KiCad), microcontroller-based development, and IoT hardware projects. Skilled in schematic design, PCB layout, BOM preparation, and hardware debugging. Experienced with ARM7, Arduino, Raspberry Pi Pico, and ESP8266 with strong knowledge of UART, SPI, and I2C protocols.

TECHNICAL SKILLS

PCB & EDA Tools	KiCad (Schematic + PCB Layout), Proteus (Circuit Simulation)
Microcontrollers	ARM7 LPC2148, ATmega328P (Arduino Uno), RP2040 (Raspberry Pi Pico), ESP8266
Protocols	UART, SPI, I2C, Bluetooth
Programming	Embedded C, C++ (Basics)
Dev Tools	Keil µVision, Arduino IDE
Hardware Skills	Schematic Design, PCB Layout, BOM Preparation, Circuit Debugging, Sensor Integration

PROJECTS

Smart Trolley System – QR & RFID Based Billing | Final Year Project

- Microcontroller-based smart cart with RFID product identification and automated billing mechanism.
- Bluetooth (HC-05) communication for real-time interaction, monitoring, and motor control for navigation.
- Complete hardware-software integration, functional testing, and debugging.

Soil Nutrient Monitoring System | IoT Hardware Project

- Sensor interface for N, P, K measurement with I2C/UART data acquisition and ESP8266 wireless transmission.

Smart Temperature & Humidity Monitoring System | Embedded Project

- Real-time monitoring using DHT11/DHT22 with threshold-based alert and display integration.

Snake Game on LED Matrix | Mini Project

- Microcontroller-based game logic with LED matrix display and joystick directional input handling.

PCB Design Projects (KiCad)

12V AC to DC Rectifier: Full-wave bridge rectifier schematic & PCB in KiCad; applied trace width, grounding, validated via Proteus.

Transformerless Power Supply: Capacitor-based supply with surge protection, zener clamping, and mains-level safety clearances in KiCad.

Servo Motor Tester (NE555 Timer): NE555 PWM generator with adjustable duty cycle for servo testing; optimized signal integrity and compact layout.

CERTIFICATIONS & TRAINING

Advanced Embedded Systems Training | Vector India Pvt. Ltd.

- Hands-on training in C programming, ARM7 (LPC2148), peripheral interfacing, and real-time embedded system design.

Embedded Systems Certification | NIELIT

- Certified in embedded system fundamentals, microcontroller programming, and hardware design.

Internship – Communication Systems | All India Radio Station

- Exposure to signal transmission, broadcast engineering, and communication hardware fundamentals.

EDUCATION

B.Tech – ECE Dr. Lankapalli Bullayya College of Engineering	2021–2025	76%
Intermediate (XII) Sri Viswa Junior College	2019–2021	87%
SSC (X) Tiny Tots High School	2018–2019	90%

EXTRACURRICULAR

- Represented university in South Zone Inter-University Basketball Championship — 3-time participant.
- Played for Visakhapatnam Port Trust basketball team; competed at district-level tournaments.